

Determinarea valorilor pime cu ajutorul metodelor numerice.

$$A \in M_n(\mathbb{C}) \quad P(\lambda) = \det(A - \lambda I), \quad \lambda \in \text{Spec } A : \exists X \neq 0, AX = \lambda X$$

$$P(\lambda) = (-1)^n \lambda^n + p_1 \lambda^{n-1} + p_2 \lambda^{n-2} + \dots + p_n$$

$$\text{Tr}(A) = \sum_{i=1}^n \lambda_i$$

$$\text{Tr}(A^k) = \sum_{i=1}^n \lambda_i^k$$

1) Metoda lui Krâlov

→ se folosește pt matrice "rare"

$$\text{Fie } A \in M_n(\mathbb{C})$$

$$y^{(0)} \in \mathbb{C}^n$$

$$P(A) = 0 \quad (\text{+ matrice } 0 \text{ verifică propriul polinom caracteristic})$$

$$\Rightarrow (-1)^n A^n + p_1 A^{n-1} + \dots + p_{n-1} A + p_n I = 0_n$$

$$y^{(p)} = A y^{(p-1)}$$

$$, p=1, 2, \dots$$

$$y^1 = A y^0$$

$$y^2 = A y^1 = A^2 y^0$$

$$y^p = A^p y^0$$

$$y^p = (y_1^p, y_2^p, \dots, y_n^p)^T$$

$$P(A)y^0 = 0$$

$$p_1 A^{n-1} y^0 + p_2 A^{n-2} y^0 + \dots + p_{n-1} A y^0 + p_n y^0 = (-1)^n A^n y^0$$

$$p_1 y^{n-1} + p_2 y^{n-2} + \dots + p_{n-1} y^1 + p_n y^0 = (-1)^n y^n$$

$$p_1 y_i^{n-1} + p_2 y_i^{n-2} + \dots + p_{n-1} y_i^1 + p_n y_i^0 = (-1)^{n+1} y_i^n, \quad i = \overline{1, n}$$

$$\Rightarrow p_i$$

2) Metoda Leverrier

$$S_k = \sum_{i=1}^n \lambda_i^k$$

$$\nabla_k = \sum \lambda_1 \lambda_2 \dots \lambda_k$$

$\binom{n}{k}$ - termenii în sumă

Relațiile lui Newton,

$$S_k - \nabla_1 S_{k-1} + \nabla_2 S_{k-2} + \dots + (-1)^k k \nabla_k = 0$$

$$S_3 - \nabla_1 S_2 + \nabla_2 S_1 - 3 \nabla_3 = 0$$

$$(-1) \nabla_1 S_2 = (\lambda_1 + \dots + \lambda_n) (\lambda_1^2 + \dots + \lambda_n^2) = S_3 + \sum \lambda_1^2 \lambda_2$$

$$(-1)^2 \nabla_2 S_1 = \sum \lambda_1^2 \lambda_2 + 3 \nabla_3$$

$$(+) \Rightarrow -\nabla_1 S_2 + \nabla_2 S_1 = -S_3 + 3 \nabla_3 //$$

$$(-1) \nabla_1 S_{k-1} = S_k + \sum \lambda_1^{k-1} \lambda_2$$

$$(-1)^2 \nabla_2 S_{k-1} = \sum \lambda_1^{k-2} \lambda_2 + \sum \lambda_1^{k-2} \lambda_2 \lambda_3 + \dots$$

$$(-1)^{k-1} \nabla_{k-1} S_1 = \sum \lambda_1^2 \lambda_2 \dots \lambda_k + k \nabla_k$$

(+) $\Rightarrow$ metoda lui Newton.

$$S_k + \nabla_1 S_{k-1} + \nabla_2 S_{k-2} + \dots + k \nabla_k = 0$$

$\rightarrow$ folosit pt determinarea polinomului caracteristic

$$A \in M_n : \exists |\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \dots \geq |\lambda_n|$$

- vom să găsim raza spectrală (λ_1)

Fie $y^{(0)} \in \mathbb{C}^n$ vector oarecare

$$y^{(0)} = \sum_{i=1}^n y_i^0 e^i$$

$$e^i = (0, \dots, 0, \underset{i}{1}, 0, \dots, 0)^T \text{ (e indice } i)$$

$$\begin{aligned} x^n - \nabla_1 x^{n-1} + \nabla_2 x^{n-2} + \dots + (-1)^n \nabla_n &= 0 \\ + | x^n - \nabla_1 x^{n-1} + \nabla_2 x^{n-2} + \dots + (-1)^n \nabla_n &= 0 \\ \downarrow \\ S_n - \nabla_1 S_{n-1} + \nabla_2 S_{n-2} + \dots + (-1)^n \nabla_n &= 0 \end{aligned}$$

$$x^1, x^2, \dots, x^m$$

$$y^{(0)} = \sum_{k=1}^m c_k x^k$$

$$y^{(p)} = A y^{(p-1)}$$

$$y^{(p)} = A^p y^{(0)} = A^p \left(\sum_{k=1}^m c_k x^k \right) = \sum_{k=1}^m c_k \lambda_k^p x^k$$

$$y_i^{(p)} = \sum_{k=1}^m c_k \lambda_k^p x_i^k$$

$$y_i^{(p+1)} = \sum_{k=1}^m c_k \lambda_k^{p+1} x_i^k$$

$$z_p = \frac{y_p^{(p+1)}}{y_i^{(p)}} = \frac{\sum_{k=1}^m c_k \lambda_k^{p+1} x_p^k}{\sum_{k=1}^m c_k \lambda_k^p x_i^k}$$

$$\lim_{p \rightarrow \infty} z_p = \lambda_1$$

$$|z_{p+1} - z_p| < \varepsilon \quad \text{condiție de oprire}$$

$$\frac{z_1 + z_2 + \dots + z_p}{p}$$